

DNS Walkthrough: From Browser Bar to Your Website

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Date: August 2026
Context: FlyRank AI Internship — PF-04 Assignment

1. What is DNS?

DNS stands for **Domain Name System**. It is the phonebook of the internet. Humans remember names like `ahmed.flyrank.ai`, but computers talk in numbers (IP addresses like `192.0.2.1`). DNS translates the human-friendly name into the machine-friendly address so the browser knows where to go.

Without DNS, you would have to memorize a string of numbers for every website you visit.

2. What is a CNAME Record?

A **CNAME** (Canonical Name) record is a type of DNS record that maps one domain name to another. Instead of pointing directly to an IP address, it says: *“This name is just an alias for that other name.”*

Think of it like a forwarding address. If someone sends mail to your old apartment, the post office forwards it to your new one. The CNAME does the same for web traffic.

My CNAME Record

When FlyRank provisions my subdomain `ahmed.flyrank.ai`, the CNAME record will look like this:

Record Type	Name (Host)	Value (Points to)
CNAME	ahmed	my-site.netlify.app

This tells the world: *“If someone asks for `ahmed.flyrank.ai`, send them to `my-site.netlify.app` instead.”*

Important: The CNAME value must be the exact Netlify site URL (the one I renamed to something clean like `ahmedelgharably.netlify.app`), not the raw IP address.

3. The Full Journey: What Happens When Someone Types Your Address

Here is the step-by-step path from typing `ahmed.flyrank.ai` to seeing the page:

Step 1: The User Types the URL

The visitor opens their browser and types `https://ahmed.flyrank.ai`.

Step 2: The Recursive Resolver

The browser does not know the IP yet. It asks a **recursive resolver** (usually provided by the ISP or a public service like Google DNS `8.8.8.8` or Cloudflare `1.1.1.1`). This resolver is like a librarian who either knows the answer or knows where to find it.

Step 3: The Root Nameserver

If the resolver does not have the answer cached, it starts at the top: the **root nameservers**. These 13 logical servers know where all the top-level domains (like `.ai`) live. The root replies: *“I do not know `ahmed.flyrank.ai`, but I know who manages `.ai`. Ask this nameserver.”*

Step 4: The TLD Nameserver

The resolver now asks the **TLD (Top-Level Domain) nameserver** for `.ai`. This server manages every domain ending in `.ai`. It replies: *“I know `flyrank.ai` is managed by these nameservers.”*

Step 5: The Authoritative Nameserver

The resolver finally asks the **authoritative nameserver** for `flyrank.ai` — this is where FlyRank Ops manages DNS records. It looks up the CNAME record for `ahmed` and replies: *“`ahmed.flyrank.ai` is an alias for `my-site.netlify.app`.”*

Step 6: Resolving the Alias

The resolver now repeats the process for `my-site.netlify.app`, eventually getting the actual IP address from Netlify’s infrastructure.

Step 7: The Response

The resolver returns the final IP address to the user’s browser. The browser opens a secure HTTPS connection to that IP, Netlify serves the website files, and the page loads with the green padlock.

4. Why HTTPS Just Works

Netlify automatically provisions an **SSL/TLS certificate** (via Let’s Encrypt) for any custom domain added to a site. When the CNAME is live and Netlify detects `ahmed.flyrank.ai` pointing to it, it requests a certificate for that domain. The browser then encrypts all traffic between the user and the server. The padlock appears without me writing a single line of certificate code.

5. Checklist for Capstone Day

When my subdomain is provisioned, I will run through this checklist:

- FlyRank Ops confirms the CNAME record `ahmed.flyrank.ai` → `my-site.netlify.app` is created on their end
- I log into Netlify → Site configuration → Domain management → Add custom domain
- I enter `ahmed.flyrank.ai` and confirm
- I wait for DNS propagation (can take minutes to 24 hours)
- I test the URL in a private/incognito window
- I confirm the HTTPS padlock is visible
- I update the link on my LinkedIn profile and CV to the new subdomain

Summary

Component	Role
Recursive Resolver	The librarian that looks up the answer for the browser
Root Nameserver	Points to the TLD nameserver
TLD Nameserver	Points to the authoritative nameserver for the domain
Authoritative Nameserver	Holds the actual DNS records (including my CNAME)
CNAME Record	An alias: <code>ahmed.flyrank.ai</code> → <code>my-site.netlify.app</code>
Netlify	Hosts the files and serves them over HTTPS

A custom domain is not a migration — it is just a pointer. The website files, the build, and the content stay exactly where they are. Only the front door changes.